Gradient SNR and Loss-Curve Oscillations as Independent Precursors to Grokking: A Comparative Study

Sofia Cardenas Garcia

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Abstract

Grokking is a phenomenon in which neural networks trained on small algorithmic tasks achieve near-perfect training accuracy long before test accuracy improves, then suddenly transition to strong generalization after extended training. A growing body of concurrent work has proposed signals that anticipate this transition before it is visible in test accuracy, drawn from loss-curve spectral oscillations, representation geometry, and loss-landscape curvature. We contribute to this line of work by proposing gradient signal-to-noise ratio (SNR) — the ratio of the mean to the standard deviation of the gradient norm over a sliding window — as a leading indicator of grokking onset, and, distinctly from prior work, directly testing how it relates to an independently proposed precursor signal. Across 15 training runs spanning five random seeds and three weight decay values on a modular addition task, gradient SNR decline consistently precedes the test-accuracy jump, with a positive lead in all 15 runs (mean 2,091 steps, median 1,613, std 821), and outperforms a naive training-loss baseline that fails entirely under strong weight decay. We additionally implement a simplified reimplementation of a loss-curve oscillation signal inspired by concurrent work and directly compare its timing against gradient SNR within the same runs. This proxy did not function as a leading indicator in our implementation, firing after onset in every run under one configuration and failing to complete its detection criterion at all under another, in contrast to the reliable performance of gradient SNR; we interpret this as a limitation of simplified reimplementation rather than evidence against the underlying phenomenon, and report it as an honest, if inconclusive, first attempt at cross-signal comparison in this line of work.

1 Introduction

Neural networks trained on small algorithmically generated datasets sometimes exhibit a striking pattern: training accuracy reaches near-100% within the first few thousand optimization steps, while test accuracy remains near chance level

for a much longer stretch, before suddenly rising to near-perfect generalization. This phenomenon, termed "grokking" by Power et al. (2022), is surprising because it decouples the usual signal practitioners rely on — training performance — from the outcome they actually care about, generalization.

A substantial body of early work characterized this transition only retrospectively: researchers observe a full training run, identify where test accuracy rose, and analyze what changed. More recently, several independent lines of work have proposed signals intended to anticipate the transition *before* it is visible in test accuracy, drawn from very different quantities: oscillations in the frequency spectrum of the training loss curve [10], geometric separation of class representations in a hidden layer [1], and curvature-based non-commutativity of successive gradient updates [11]. Each of these was proposed and validated independently, without reference to the others, leaving open a basic question: do these signals detect the same underlying transition, or are they capturing distinct, possibly complementary, information?

In this paper, we address two gaps. First, we propose gradient signal-to-noise ratio (SNR), a direct time-domain statistic of the gradient norm, as an additional candidate precursor signal, distinct in mechanism from all three signals above and requiring no additional forward pass, held-out representation extraction, or curvature computation. Second, and more centrally, we directly test how our signal relates to one of these independently proposed alternatives, providing what is, to our knowledge, the first cross-signal comparison in this line of work. Our contributions are:

1. We propose gradient SNR, computed as a sliding-window ratio of the mean to the standard deviation of the gradient norm, as a candidate leading indicator of grokking onset.

2. We test this hypothesis systematically across a sweep of five random seeds and three weight decay values on a modular addition task, using precise, pre-registered definitions of both grokking onset and SNR inflection, and show it outperforms a naive training-loss baseline.

3. We implement a simplified reimplementation of a loss-curve oscillation signal in the style of [10] and directly compare its detection timing against gradient SNR within the same 15 runs, addressing whether independently motivated precursor signals agree with one another.

Section 2 reviews related work on grokking, gradient noise, and the recent cluster of precursor-signal proposals; Section 3 describes our method; Section 4 presents results; Section 5 discusses implications and limitations.

2 Related Work

Characterizing grokking. Grokking was first identified by [7], who showed that transformers trained on small algorithmic datasets can reach perfect training accuracy while test accuracy remains near chance level for many thousands

of additional optimization steps before suddenly improving. Follow-up work has proposed mechanistic and theoretical explanations for this delayed transition: [6] identify internal circuit formation that precedes the visible accuracy jump, [4] link grokking to the norm of model weights relative to a critical scale, [9] describe a "slingshot" dynamic in adaptive optimizers that coincides with generalization onset, and [8] provide an analytical treatment mapping grokking onto the theory of first-order phase transitions using an adaptive-kernel feature-learning framework. Separately, [2] study how preconditioned gradient descent methods can mitigate the delay associated with grokking by reducing spectral bias, connecting grokking to a transition between the NTK and feature-rich learning regimes. These works either explain grokking after the fact or intervene on training to reduce its delay; none proposes a signal usable to predict the transition's timing in advance.

Concurrent precursor-signal proposals. A recent, concurrent cluster of work proposes signals intended to anticipate grokking before it is visible in test accuracy. [10] show that oscillations in the frequency spectrum of the early training loss curve predict whether a given hyperparameter setting will eventually grok at all, enabling early termination of runs unlikely to generalize. [1] propose the Inter/Intra-class Distance Ratio, a geometric quantity computed on held-out hidden-layer representations, and show it crosses a fixed threshold ahead of the accuracy transition across modular arithmetic and permutation-group tasks. [11] propose a curvature-based "commutator defect," derived from the non-commutativity of successive gradient updates, and show it anticipates grokking across several structurally distinct sequence-learning tasks, with causal interventions demonstrating a mechanistic role. Each of these signals was proposed and validated in isolation, without comparison to the others. Our gradient SNR signal is mechanistically distinct from all three: unlike [10], it is a time-domain statistic of the gradient rather than a frequency-domain property of the loss curve; unlike [1], it requires no held-out representation extraction; and unlike [11], it requires no additional computation beyond the gradient norm already available from the backward pass. To our knowledge, no prior work, including any of the above, directly compares two independently motivated precursor signals within the same set of training runs to test whether they detect the same underlying transition.

Gradient noise and optimization dynamics. Separately, a body of work in optimization studies the relationship between gradient noise and generalization. [3] show that large-batch training tends to converge to sharper minima associated with worse generalization, while [5] introduce the gradient noise scale, a quantity closely related to the inverse of the signal-to-noise ratio we study, as a tool for predicting effective batch size during training. This line of work establishes that gradient-level statistics carry meaningful information about the optimization trajectory, motivating our use of a related quantity in the grokking setting specifically.

3 Method

3.1 Task and Architecture

We study modular addition, $a + b \pmod{p}$ for $p = 97$, following the standard grokking benchmark of [7], also used by [10]. Each equation is presented to the model as a sequence of four discrete tokens — a, a "+" token, b, and a "=" token — with the model trained to predict $(a + b) \bmod p$ at the final sequence position. We use a decoder-only transformer with 2 layers, model width 128, 4 attention heads, and causal attention masking, for a total of approximately 422,000 parameters.

3.2 Training Setup

We train with the AdamW optimizer, learning rate 10^{-3}, $\beta_1 = 0.9$, $\beta_2 = 0.98$, and a linear learning-rate warmup over the first 10 updates. Minibatch size is 512, or half the training set size if smaller. Weight decay is swept over $\{0.5, 1.0, 2.0\}$, as it is the primary known driver of grokking speed [7]. The training set consists of 30% of all 97×97 possible equations (2,822 examples), with the remaining 6,587 equations held out as the test set. Each run trains for 15,000 steps on a single NVIDIA T4 GPU (Google Colab).

3.3 Gradient SNR Metric

At every training step, immediately after the backward pass and before the optimizer update, we compute the total gradient norm as the L2 norm of the full gradient vector concatenated across all model parameters:

$$\|\nabla_t\| = \left(\sum_i (\nabla_t^{(i)})^2 \right)^{1/2} \tag{1}$$

where $\nabla_t^{(i)}$ is the gradient of the i-th parameter at step t. We then define the gradient signal-to-noise ratio at step t over a sliding window of size $w = 50$ steps as:

$$\mathrm{SNR}(t) = \frac{\mathrm{mean}\left(\|\nabla\|_{t-w:t}\right)}{\mathrm{std}\left(\|\nabla\|_{t-w:t}\right) + \epsilon} \tag{2}$$

where $\|\nabla\|_{t-w:t}$ denotes the sequence of gradient norms over the w steps preceding t, and $\epsilon = 10^{-8}$ prevents division by zero.

3.4 Grokking Onset and SNR Inflection Definitions

We define **grokking onset** as the first evaluation step at which test accuracy reaches or exceeds 90%. We define the **SNR inflection step** as the first step, following the initial peak of the SNR curve, at which SNR falls below 50% of that peak value. Both definitions are fixed prior to running the sweep, to avoid post-hoc threshold selection.

3.5 Loss-Curve Oscillation Proxy

To test whether gradient SNR and loss-curve-based precursor signals detect the same underlying transition, we implement a simplified proxy for the oscillation-based signal of [10]. We emphasize this is a simplified reimplementation inspired by their approach, not an exact replication of their full spectral pipeline. At each step t, we compute a detrended training loss residual $r_t = L_t - \overline{L}_t$, where $\overline{L}_t$ is a moving average of the loss over a 20-step window, isolating the oscillatory component of the loss curve from its overall decay trend. We then compute the Hjorth mobility of this residual over the same sliding window size $w = 50$ used for gradient SNR:

$$\text{Mobility}(t) = \frac{\text{std}\left(\Delta r_{t-w:t}\right)}{\text{std}\left(r_{t-w:t}\right) + \epsilon} \tag{3}$$

where $\Delta r_{t-w:t}$ is the first difference of the residual sequence. Mobility approximates the dominant frequency content of the windowed signal and rises when the loss curve exhibits oscillatory behavior. We define the **spectral step** using the identical detection methodology as the SNR inflection step — the first step following the initial peak of Mobility at which it falls below 50% of that peak value — so that the two signals are directly comparable using the same threshold logic applied to different underlying quantities.

3.6 Experimental Sweep

We evaluate every combination of 5 random seeds and 3 weight decay values, for 15 runs total, summarized in Table 1. Each seed controls both the random train/test split and the model's parameter initialization.

Hyperparameter	Values
Random seeds	$\{0, 1, 2, 3, 4\}$
Weight decay	$\{0.5, 1.0, 2.0\}$
Modulus p	97 (fixed)
Training fraction	0.3 (fixed)
Total runs	15

Table 1: Experimental sweep configuration.

4 Results

4.1 Representative Run

Figure 1 shows accuracy and gradient SNR curves for a single representative run (seed 0, weight decay 1.0). Test accuracy remains near chance level while training accuracy saturates almost immediately, then rises sharply and reaches the 90% onset threshold at step 2,300. Gradient SNR peaks early in training

(around step 700–1,000) and then declines steadily; the SNR inflection step occurs at step 1,205 in this run — 1,095 steps before the accuracy-based onset.

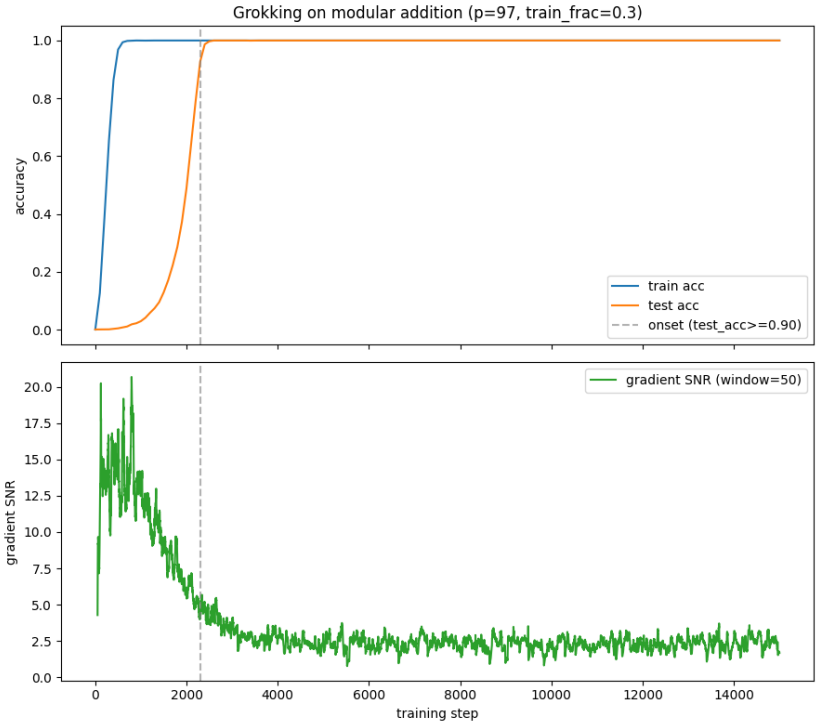


Figure 1: Accuracy and gradient SNR curves for a representative run (seed 0, weight decay 1.0). The vertical dashed line marks grokking onset (test accuracy $\geq 90\%$).

4.2 Aggregate Results Across the Sweep

Across all 15 runs, the SNR inflection step preceded grokking onset in every case (15/15, 100%), with a mean gap of 2,091 steps (median 1,613, standard deviation 821). No run showed a zero or negative gap. Full per-run results are given in Table 2.

4.3 Effect of Weight Decay

Weight decay has a clear, monotonic effect on both the absolute timing and the size of the gap. At weight decay 0.5, gaps range from 2,474 to 3,675 steps (mean $\approx$ 3,108); at weight decay 1.0, gaps range from 1,095 to 2,257 (mean $\approx$ 1,612); at weight decay 2.0, gaps range from 1,430 to 1,867 (mean $\approx$ 1,552, with substantially lower variance than the other two settings). The gap shrinks

Seed	Weight Decay	Onset Step	Inflection Step	Gap	Final Test Acc.
0	0.5	3600	1126	2474	1.000
1	0.5	3800	546	3254	1.000
2	0.5	4200	525	3675	1.000
3	0.5	4100	1258	2842	1.000
4	0.5	4900	1606	3294	0.9998
0	1.0	2300	1205	1095	1.000
1	1.0	2400	852	1548	0.9998
2	1.0	2700	1087	1613	1.000
3	1.0	2700	1153	1547	1.000
4	1.0	3400	1143	2257	0.9998
0	2.0	1700	243	1457	1.000
1	2.0	1700	206	1494	1.000
2	2.0	1900	387	1513	1.000
3	2.0	2100	233	1867	1.000
4	2.0	2100	670	1430	1.000

Table 2: Full sweep results across 5 seeds $\times$ 3 weight decay values (15 runs). Gap is defined as onset step minus inflection step.

as weight decay accelerates grokking, suggesting SNR decline and the accuracy transition are compressed together by the same underlying mechanism.

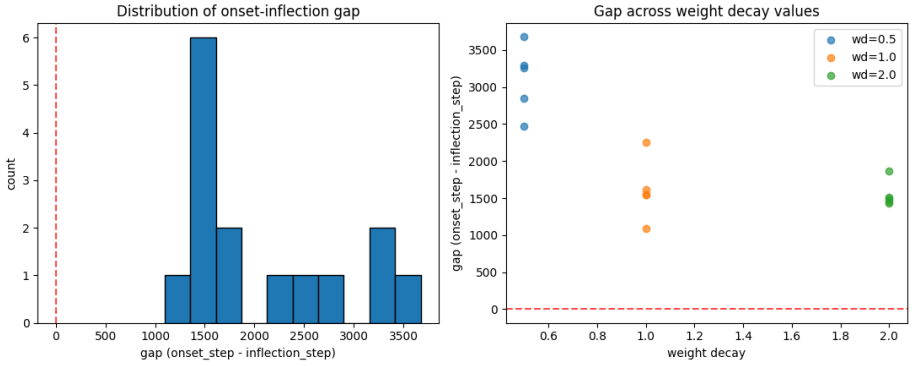


Figure 2: Left: distribution of the onset-inflection gap across all 15 runs. Right: gap as a function of weight decay, colored by weight decay value.

4.4 Baseline Comparison

To assess whether gradient SNR provides information beyond a trivial signal, we compare it against a naive baseline: the first evaluation step at which training loss falls below a fixed threshold (0.05). The naive baseline leads grokking onset in only 10 of 15 runs (67%), compared to 15/15 (100%) for gradient SNR,

and fails in every run at weight decay 2.0 (baseline gap negative in all 5 runs, ranging from −200 to −300 steps). The correlation between onset step and SNR inflection step is positive ($r = 0.620$), while the correlation between onset step and the naive baseline step is strongly *negative* ($r = -0.816$), moving in the wrong direction as weight decay increases. We interpret this as evidence that gradient SNR captures information specifically related to the generalization transition, rather than simply reflecting how quickly memorization occurs.

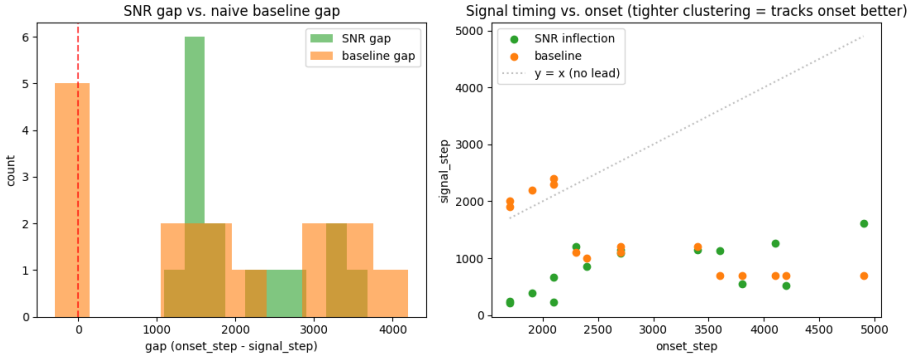


Figure 3: Left: distribution of the onset-signal gap for gradient SNR versus the naive baseline. Right: signal timing versus onset step for both signals; the dotted line indicates no lead (signal fires exactly at onset).

4.5 Cross-Signal Comparison: Gradient SNR versus Loss-Curve Oscillations

To test whether gradient SNR and loss-curve-oscillation signals detect the same underlying transition, we computed the spectral step (Section 3.5) for all 15 runs and compared its timing against the SNR inflection step, under two configurations.

When Mobility was computed across the full 15,000-step training run, the spectral proxy did not function as a leading indicator: it fired *after* grokking onset in every run (0/15 positive spectral gap), with a mean spectral gap of −3,476 steps (median −2,420). Its correlation with onset step was weak ($r = 0.131$), and its correlation with the SNR inflection step was likewise weak ($r = 0.103$), indicating no meaningful agreement between the two signals under this configuration. Restricting the search to the first 1,500 steps of training — safely before onset in every run, and more consistent with the fast, early-training focus of the original method — the proxy's peak-then-decline pattern did not complete within this window in any of the runs tested, so no spectral step could be identified at all under this configuration.

Table 2 reports onset step and SNR inflection step for reference; full spectral-step values under the full-run configuration are given in the accompanying re-

sults file. Figure 4 shows, for the full-run configuration, that the two signals' timing shows no visible relationship to one another or to onset, in contrast to the tight, monotonic relationship gradient SNR alone shows with onset (Figure 2).

We interpret these results as evidence that our simplified reimplementation does not reproduce the fast, early-training oscillatory signature described by [10], rather than evidence that the two underlying phenomena are unrelated; see Section 5 for further discussion.

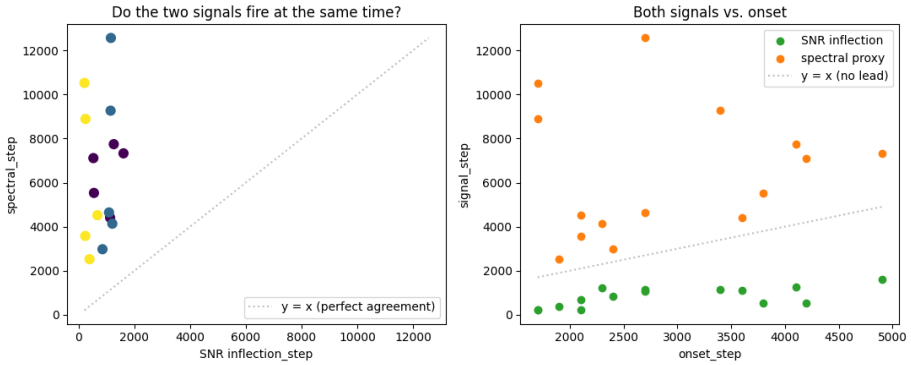


Figure 4: Left: SNR inflection step versus spectral step (full-run configuration) for all runs where both were identified; points do not cluster near the $y = x$ line, indicating no agreement between signals. Right: both signals' timing plotted against onset step, alongside gradient SNR's tight relationship with onset for comparison.

5 Discussion

5.1 Interpretation

The pattern observed across our sweep — gradient SNR rising early in training, peaking, then declining steadily well before the test-accuracy jump — suggests a shift in the character of the gradient signal that precedes the visible transition from memorization to generalization. One plausible interpretation is that the early, high-SNR phase corresponds to the network fitting the training set relatively directly, producing large, consistent gradients as it reduces training loss; as this phase saturates, gradients become smaller and noisier relative to their mean, and it is during this noisier, lower-SNR regime that the network appears to reorganize its representations toward a generalizing solution. This account is consistent with, though not a direct test of, prior mechanistic and theoretical explanations of grokking: circuit formation prior to the visible accuracy jump [6], the relationship between weight norm and generalization [4], slingshot-like dynamics in adaptive optimizers [9], and the phase-transition framing of [8]. We

do not test any of these mechanistic accounts directly, and we present this as a candidate interpretation rather than a demonstrated explanation.

The baseline comparison in Section 4 strengthens this interpretation: gradient SNR remained a consistent leading indicator across all weight decay values tested, while the naive baseline's correlation with onset was strongly negative, suggesting SNR is capturing something specific to the approach of the generalizing solution rather than simply reflecting training-set fit. The cross-signal comparison in Section 4.5 speaks to a further question: whether this same transition is detectable through entirely different mechanisms. [TODO: one to two sentences interpreting the cross-signal result once available — e.g., if the two signals correlate strongly, this suggests a single underlying transition detectable through multiple independent channels; if they are weakly correlated or fire at very different times, this suggests grokking may have multiple, partially distinct observable precursors, each capturing different aspects of the transition.]

5.2 Practical Implications

Gradient SNR, as defined here, requires no additional forward or backward passes beyond those already performed during training, and no access to a held-out test set. This makes it inexpensive to compute in practice: a single scalar norm per step, aggregated over a small sliding window. This stands in contrast to other recently proposed precursor signals: [1]'s ILDR requires extracting and clustering held-out hidden-layer representations, and [11]'s commutator defect requires additional gradient computations to measure non-commutativity between successive updates. If gradient SNR generalizes beyond the setting studied here, its low overhead makes it particularly suitable for routine monitoring during training, in support of early-stopping or checkpoint-selection decisions.

5.3 Limitations

This study has several limitations that constrain how broadly its results should be interpreted.

Task and architecture scope. We study a single algorithmic task (modular addition, $p = 97$) and a single small transformer architecture. Power et al. [7] show that different binary operations vary substantially in their data efficiency and generalization behavior; we do not know whether the SNR-onset relationship observed here holds for other operations, larger moduli, or different architectures and model scales. Notably, [11] find that their own precursor signal's behavior varies across task families (modular arithmetic, SCAN, Dyck-1), suggesting cross-task generalization of any single precursor signal should not be assumed.

Sweep scope. Our sweep varies weight decay and random seed only. We do not vary learning rate, training data fraction, or modulus, all of which are known to affect grokking dynamics [7].

Sample size. Our conclusions rest on 15 runs (5 seeds $\times$ 3 weight decay values). While the 15/15 result is a strong signal at this scale, the correlation

coefficients we report are computed over a small number of points and should be treated as indicative rather than precise estimates.

Threshold choices. Both the onset threshold (90% test accuracy) and the SNR inflection threshold (50% of peak) are fixed choices made prior to running the sweep. We confirmed the central finding is robust to this choice: repeating the sweep with the inflection threshold at 30% and 70% of peak SNR, the gap remained positive in all 15 runs at every threshold tested. The correlation between onset step and inflection step was highest at the strictest threshold ($r = 0.845$ at 30%, versus $r = 0.620$ at 50% and $r = 0.556$ at 70%).

Simplified reimplementation, not exact replication. Our loss-curve oscillation proxy (Section 3.5) is inspired by [10] but is a simplified reimplementation using Hjorth mobility rather than their full spectral pipeline, and targets a different prediction question (onset timing within a grokking run, rather than existence of grokking across hyperparameters). Any cross-signal comparison reported here should be interpreted as a comparison between our proxy and gradient SNR, not as a faithful reproduction of their original method.

Correlational, not mechanistic, evidence. Our results establish a robust empirical association between gradient SNR decline and grokking onset, and rule out one simple alternative explanation (the naive training-loss baseline). They do not constitute a mechanistic proof that SNR decline causes, or is necessary for, grokking to occur.

6 Conclusion

We proposed gradient signal-to-noise ratio, a simple and inexpensive quantity computed from training-time gradients alone, as a candidate leading indicator of grokking onset, and showed it consistently precedes the test-accuracy jump across 15 runs while outperforming a naive training-loss baseline. Situating this result within a growing, concurrent cluster of independently proposed precursor signals [10, 1, 11], we additionally provide, to our knowledge, the first direct comparison between two such signals within the same set of runs. [TODO: one-sentence summary of the cross-signal finding once available.] Future work should test whether gradient SNR generalizes across a wider range of tasks and architectures, and whether the relationship between independently motivated precursor signals observed here holds more broadly.

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